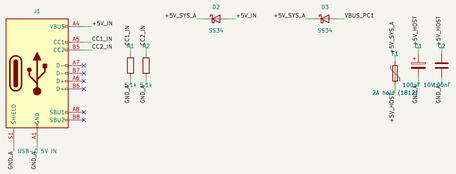
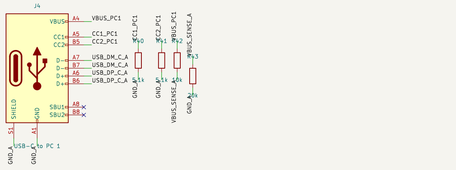
Power input — 5 V adapter (J1) OR-ed with PC 1 VBUS (D2/D3)

D2/D3 make the adapter and PC 1 share the load without ever back-feeding each other.
Works without the adapter as long as PC 1 is on; with the adapter the keyboard works while PC 1 is off.

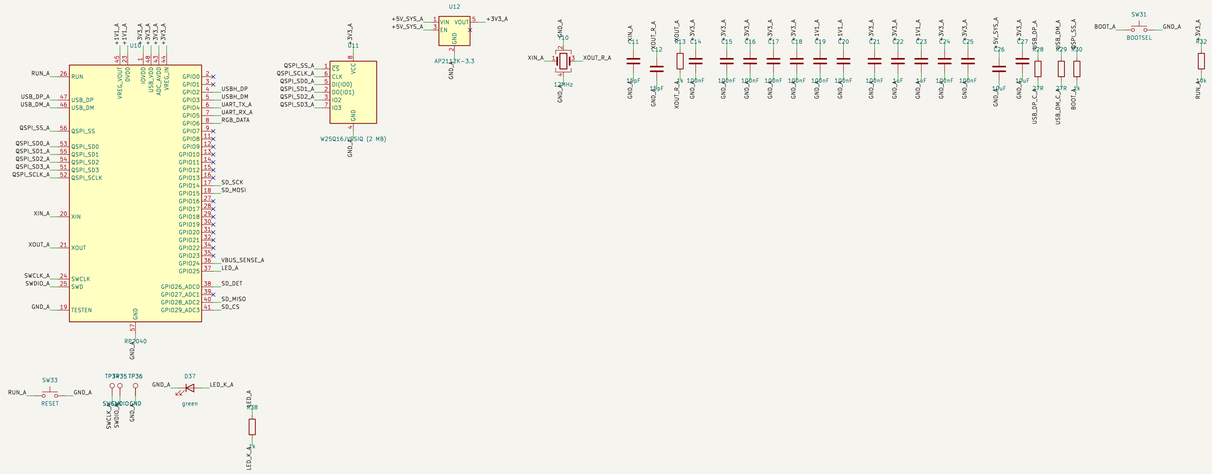


PC 1 USB-C (device) + VBUS sense on GPIO24

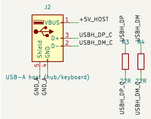


MCU-A — RP2040 (USB host, router, microSD)

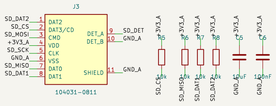
GPIO2/3 = PIO-USB host, GPIO4/5 = UART1, GPIO14/15/28/29 = SPI1 + CS, GPIO26 = card detect, GPIO25 = LED. Same pin map as variant A.



USB host port (keyboard, mouse, dongle or hub)

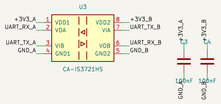


microSD — shared drive

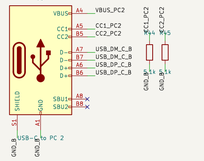


Galvanic isolation (GND_A and GND_B never meet)

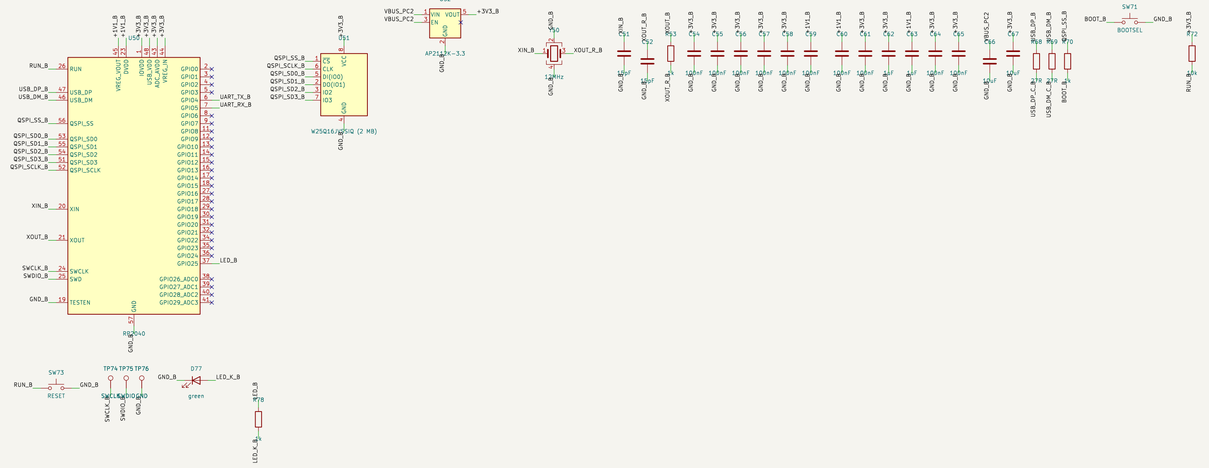
CA-IS3721HS (pin compatible with ADuM1201).



PC 2 USB-C (device, powers side B)

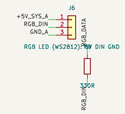


MCU-B — RP2040



Optional RGB LED header (WS2812/SK6812): 5V, DIN, GND

Data from GPIO6 through R9 (same pin as variant A).



Mechanical



Power flags

